Artificial Intelligence-Driven Malware Detection via Windows Event Log Engineering in Smart Grid Control Systems

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Agenda

Motivation

Related Work

Threat Model

Data and Features

Model

Results

Discussion

Motivation

Smart Grid Cyber Threat Landscape

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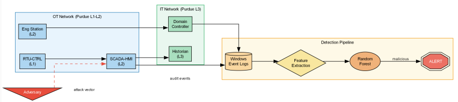
- High-profile incidents: Stuxnet [1], Ukrainian grid attacks (2015, 2016), Colonial Pipeline (2021)
- Regulatory mandates: NERC CIP-007 requires event logging; IEC 62351 specifies audit trails
- **Gap**: Windows Event Logs at the OT layer are untapped for intrusion detection



Related Work

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- **Drain** [2] — online log parsing with fixed-depth tree
- **DeepLog** [3] — LSTM sequence modelling of log keys
- **LogAnomaly** [4] — semantic vector anomaly detection

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- **LogAnomaly** [4] — semantic vector anomaly detection
- **Common limitation**: none applied to ICS/SCADA Windows hosts or ICS-specific event semantics

Related Work: ICS Intrusion Detection

- **Carcano et al.** [5] — critical-state analysis for SCADA command sequences
- **Goldenberg & Wool** [6] — Modbus/TCP protocol modelling
- **SWaT dataset** [7] — physical/network measurements, no host audit logs

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- **Mitchell & Chen** [8]: host-level IDS is rare in CPS literature
- **Our contribution**: first framework targeting Windows Event Logs at Purdue L1–L3

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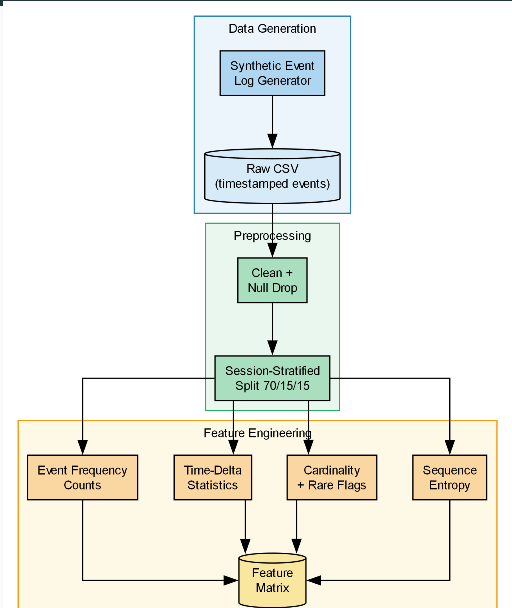
4 Attack Patterns Modelled (MITRE ATT&CK ICS)
Lateral movement · Persistence · Privilege escalation · Reconnaissance

Data and Features

Synthetic Event Log Generator



- 6,000 sessions: 5,000 benign + 1,000 malicious (5:1 ratio)
- 89,288 total events
- 13 Windows Event IDs (4624, 4625, 4663, 7045, …)

Session-Level Feature Engineering

Group	Features	Dim.
Event frequency counts	count per event ID + total	14
Time-delta statistics	mean, std, min, max gap	4
Host/user cardinality	unique sources, users, hosts	3
Rare-event flags	binary indicators	var
Sequence entropy	Shannon H over event IDs	1

Total: 24-dimensional feature vector per session

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- Sub-millisecond inference latency per session

Training Protocol
RandomizedSearchCV (20 configs) · 5-fold stratified CV · macro F1 objective

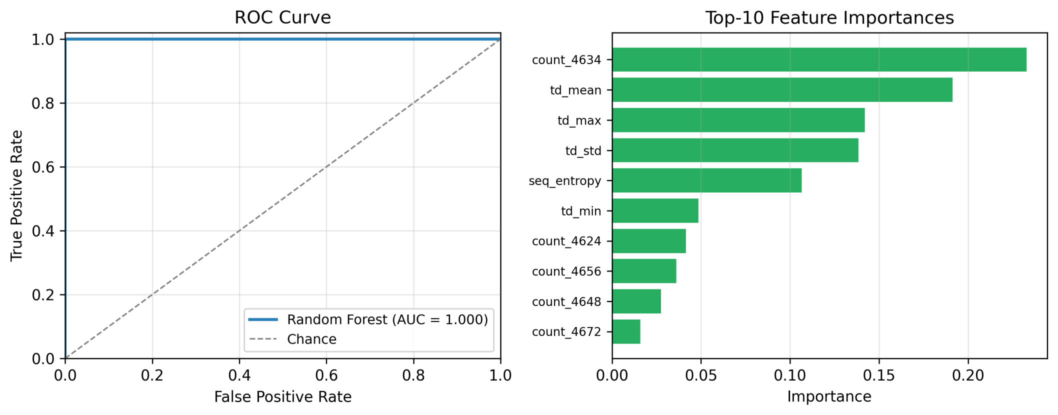
Results

Classification Results

Method	Prec.	Recall	F1 (macro)	ROC-AUC
IF (unsupervised)	0.611	0.594	0.601	0.735
LR	0.917	0.974	0.942	0.994
SVM (RBF)	0.818	0.881	0.844	0.951
Decision Tree	0.982	0.987	0.984	0.988
XGBoost	0.997	0.999	0.998	1.000
RF (proposed)	**1.000**	**1.000**	**0.999**	**1.000**

RF: mean $\pm$ std across 5 seeds $= 0.999 \pm 0.001$

ROC Curve and Feature Importances



Ablation Study

Removed Feature Group	F1 (macro)	ΔF1
None (full)	1.000	—
Group 1: Event counts	0.944	−0.056
Group 2: Time-delta stats	0.972	−0.028
Group 3: Cardinality	1.000	0.000
Group 4: Rare-event flags	1.000	0.000
Group 5: Sequence entropy	1.000	0.000

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Event counts + timing sufficient; Groups 3–5 redundant (not useless).

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- **Latency**: $<$0.5 s per 900-session batch on single CPU core — SIEM-compatible

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- First open-source framework for Windows Event Log-based malware detection in smart grid ICS
- 24-feature session pipeline; Random Forest with class-imbalance compensation
- $F1 = 0.999 \pm 0.001$ (5-seed), $\text{ROC-AUC} = 1.000$; XGBoost competitive (0.998)
- Event counts most discriminative (ablation $\Delta F1 = -0.056$); evasive attacks reduce single-feature separability from $F1 = 1.0$ to 0.86
- Framework enables reproducible ICS security research without operational data

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Future Work
Real ICS logs · Multi-class ATT&CK attribution · Online learning · Adversarial robustness against evasive attackers on real logs

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Questions

Thank you.

Questions welcome.